Monday, May 25, 2026 7:40 PM

Classification METRICS ⊛ Why? ⇒ V.Imp

- Business case
- Accuracy — Problems
- Confusion matrix → Precision, Recall, f1 score
- Cross validation →

Evaluate → metrics

⊛ Log Reg^n

Class^n metrics ⇒ ⊛ Accuracy ✗

⊛ log Reg^n → 1 ⇒

ps → gmail, outlook → 90%. —

legit email

⊛⊛⊛ spam V/s Ham



0
1
0
1

⊛ Class 1 → spam (+ve)

↳ ⊛ Class of interest

Class 0 → Non spam (-ve)

* Tabular data → Textual data

IP → Vector

[0.01, 0.50, 3.2]

* Bag of words / Tf-idf

$100 → $100 vectors

* Imbalanced data ⇒

* 72% → (Not spam)
* 28% → (spam)

⇒ Imbalanced data

(50 50

92% ⇒ good or Bad

Target

⇒ not very good

⇒

0 (80%.) 1 (20%.)

20 → +ve

80 → -ve

imbalanced

(*) 48% → +ve

52% → -ve

Balanced data

(*) Accuracy ⇒ 92%

→ not a good model

Test data → spam = 360 emails

Not spam = 840 emails

⇒ imbalanced data

1200 → NS

↓

accuracy ⇒ Bad metric

840 ✓

Dumb model

↳ predicts every email as

↳ Predicts every email as Non spams

(*) Accuracy of Dumb model = 840/1200 = 70%

Correct / Total

(*) good ↓

70% → still manageable

logical enough

X Cancer = 100 - Non cancer = 1100

cancer detection ⇒

(*) Accuracy = 1100/1200 = 91½

↳

(*) Finding life of Positives

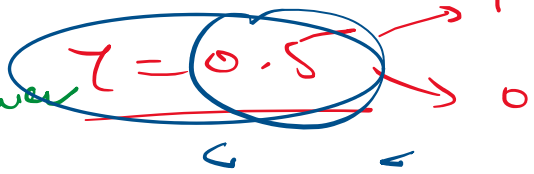
model → Biased → imBalanced data

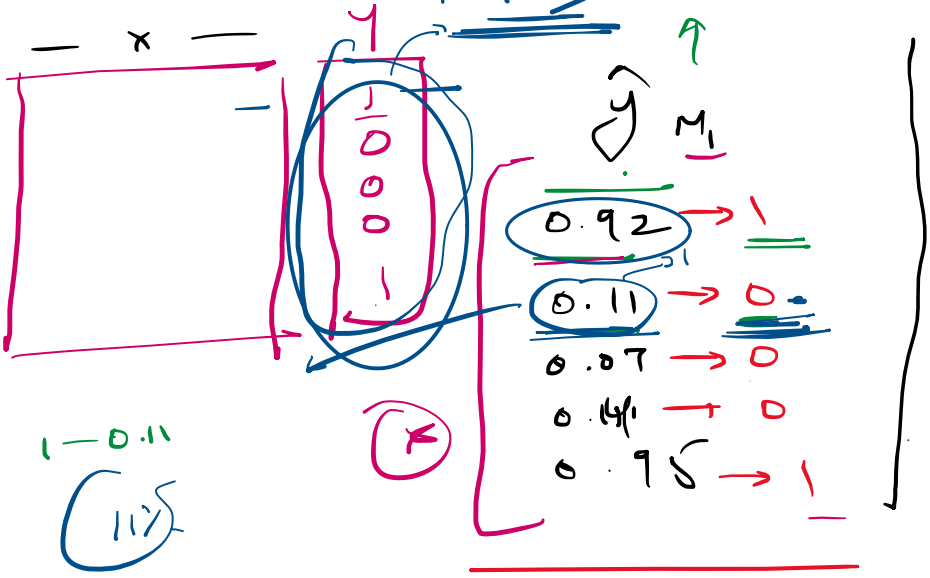
↳ DP → equal

Issues with Accuracy

1. Imbalance → Accuracy is misleading

2. Prob. is not considered ⇒





$acc_1 = 100\%$

$acc_2 = 100\%$

M_1 & M_2 → same

↳ more confidence

Quiz time!

Time Left: 3s

if data1: 20 Cancer Patients and 100 non-Cancer Patients and data2: 80 Cancer Patients 100 non-Cancer Patients, then:

Patients have participated

A	Data1 = Imbalance, Data2 = balance	66%
B	Data1 = balance, Data2 = Imbalance	14%
C	Data1 = balance, Data2 = balance	2%
D	Data1 = Imbalance, Data2 = Imbalance	18%

End Quiz Now

data 1 → 20 cancer, 100 Non Cancer

data 2 → 80, 100 ⇒

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list of metrics used to evaluate Regression model ⇒

* MSE, MAE, R^2, Adj R^2

Quiz time!

Time Left: 16s

In the evaluation of a classification model, what does the accuracy metric represent?

32 users have participated

A	The model's ability to handle imbalanced datasets	0%
B	The precision of the model in predicting positive instances	
C	The ratio of true positive predictions to the total predictions	31%
D	The overall correctness of the model's predictions across all classes	66%

End Quiz Now

* ⇒ Time

NS ↑ 0 | spar ↑ 1

* interest

* CONFUSION MATRIX ⇒

↳ correct | incorrect pred^n in

each class

← → made incorrect pred^n for spam ←
u said it is NOT spam
→ mistake

← → email → NS , By mistake → spam
→ mistake

EM

Prediction ($\hat{y}$)

2 classes
1
2×2
matrix

Actual (y) \ Prediction ($\hat{y}$)	0(NS)	1(S)
0(NS)	1 (correct)	2 (incorrect)
1(S)	3 (incorrect)	4 (correct)

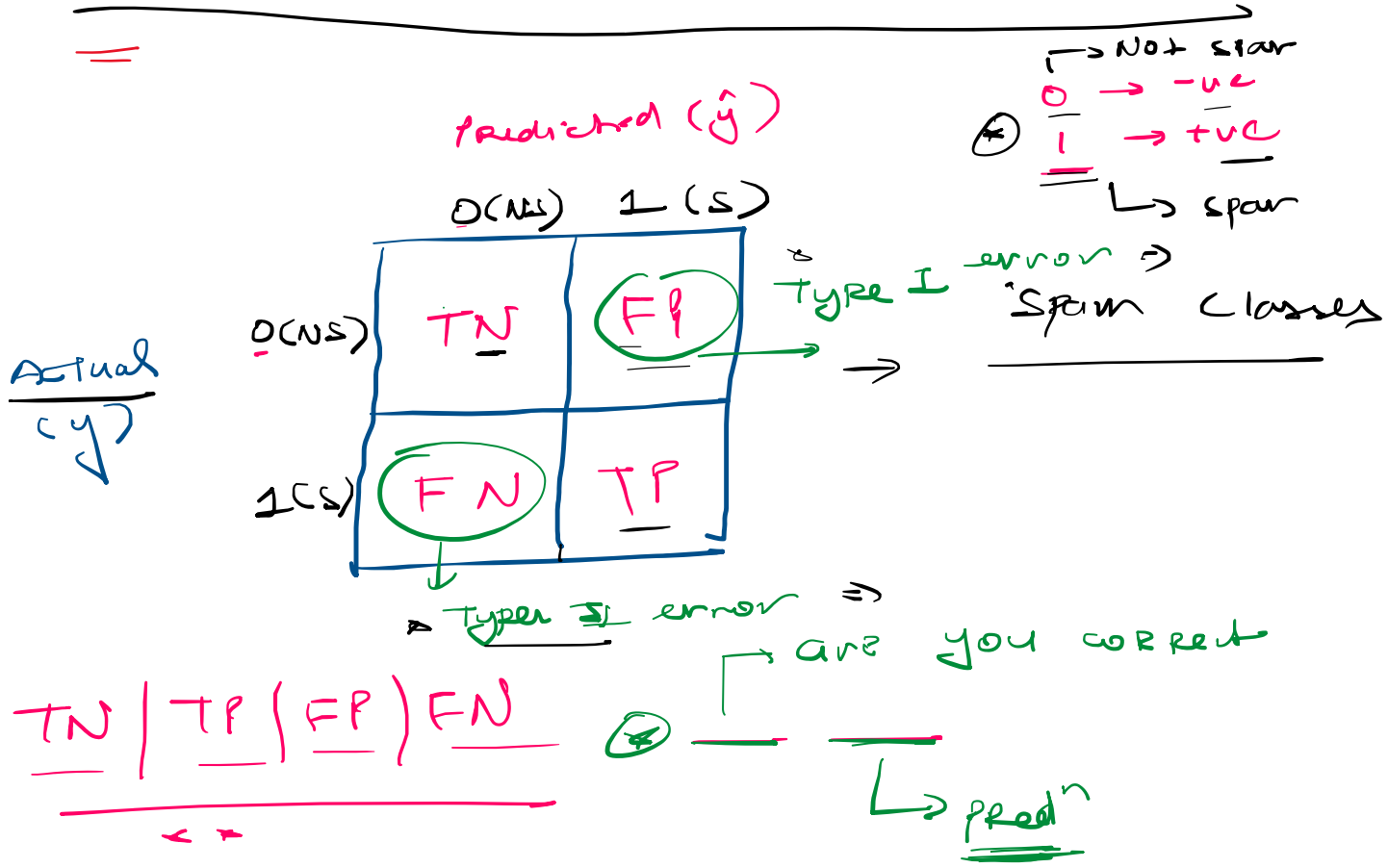
→ [3- classes
3×3]

(1) $y=0$ & $\hat{y}=0$ (correct)

(2) $y=0$ & $\hat{y}=1$ (incorrect)

(3) $y=1$ & $\hat{y}=0$ (incorrect)

(4) $y=1$ & $\hat{y}=1$ (correct)



Quiz time!

Time Left: 19s

if model classifies students into classes A,B and C , then Confusion matrix looks like ?

7 users have participated

A	12 x 12	14%
B	2 x 2	0%
C	4 x 4	0%
✓ D	3 X 3	86%

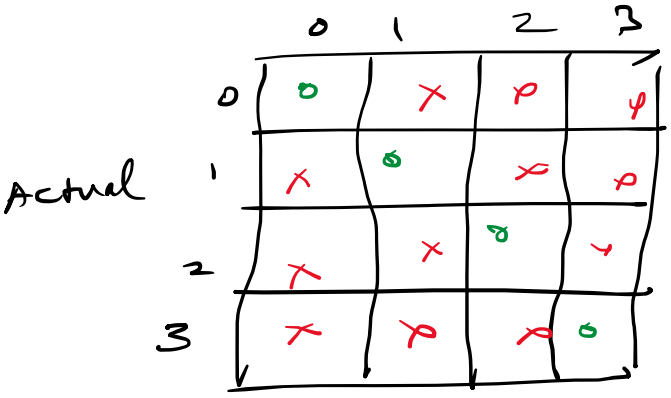
End Quiz Now

2 classes → 2×2

3×3

4 classes ⇒

Pred.



Test

⇒ 90 : 10

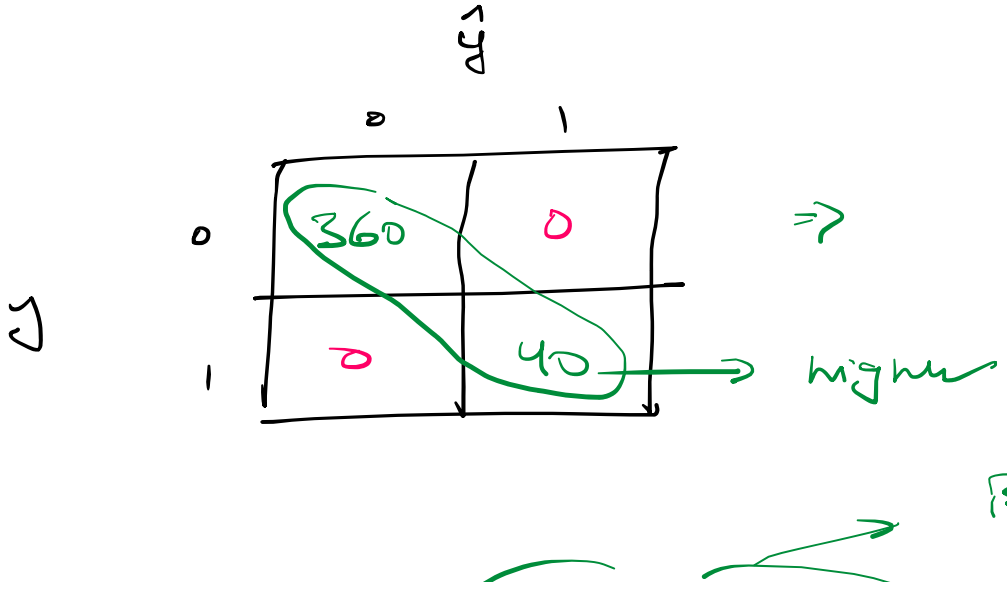
Ratio

400 Df

360 NS

40 Spam

ideal confusion matrix



Baseline model

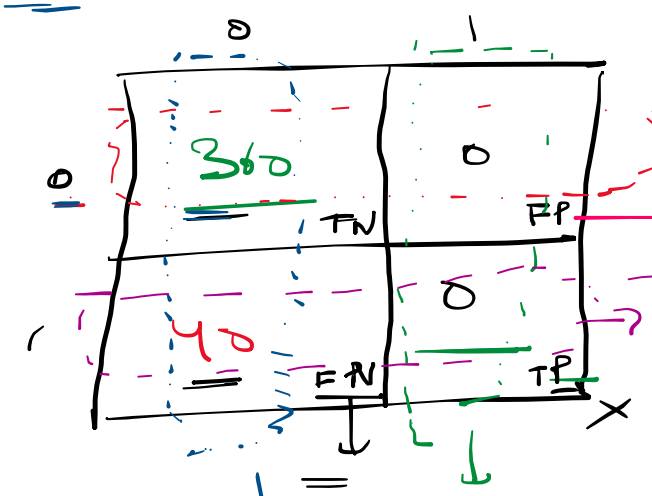
CM for DumB model ⇒ (Passive model)

400 emails → Not spam = 400 | spam(1) = 0

Actual: 360 → NS, 40 → S

360/400 = ↑ (logical)

DumB → 400



How many emails were actually Not spam that is predicted as spam

DumB model → $\hat{N}$ $\hat{P}$

Spam Vs Ham

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

5 min Break.

5 min Break.

Quiz time!

Time Left: 40s

For Ideal Model, which of the following is true?

14 users have participated

A	FP and FN ⇓, while TP and TN ⇑	79%
B	TP and TN ⇓, while FP and FN ⇑	21%
C	TP and FN ⇓, while FP and TN ⇑	0%
D	FP and TN ⇓, while TP and FN ⇑	0%

End Quiz Now

TP & TN ↑

FN

Precision

actual = 1
predr = 0

(1) Receiving a spam

inBox FN

(2) Missing out on offer letter (spam)

actual = 0
predr = 1

FP

FP

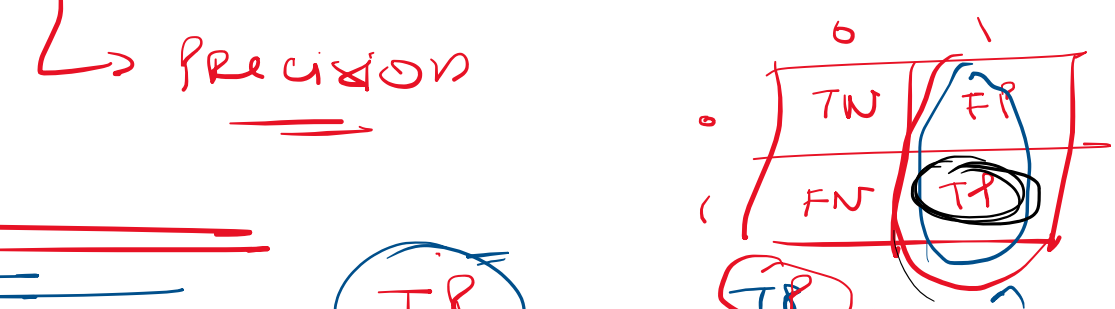
more dangerous

FP → more dangerous

TP ↑ , FP ↓

goal → TP ↑ ⇒ detect as many spam emails

↳ Precision



⊛ Precision = TP / (TP + FP) = TP / P̂

TP + FP → Total Spam Predicted

P̂

Spam V/s Ham

Spam → +ve → 1

NS → −ve → 0

eg ⇒ 80 spam emails ⇒ prediction

Actual spam = 32

(4) Precision = $\frac{32}{80}$ = 40% = 0.4

Another model

correctly pred spam = 66

(5) Precision = $\frac{66}{80}$ = 82% = 0.82

* model ⇒ Have High Precision



if FP = 0 ⇒ FP

(7) Precision = $\frac{TP}{TP + 0} = 1 = 100\%$.

Precision

(8) ⇒ how precisely you can predict the +ve class

A \ P	0	1
0	360	0
1	40	0

(9) Dumb model

Precision = $\frac{0}{0+0} = 0$

(10) Train & test

(11) ideal model

	0	1
0	360	0
1	0	40

TP

TP + FP

(12)

precision = $\frac{40}{40+0} = 1$

(13) Range of precision ⇒ 0 to 1

[0, 1]

Range of Precision → [0, 1]

Precision = Correct +ve / Total +ve

Business → High Precision

TP ↑ FP ↓

Cancer → 1 → interested

Not cancer → 0

Cancer → 1 +ve

No → -ve → 0

Problem

actual → 0, Predⁿ → 1 → FP

1. Healthy patients is considered cancerous

actual = 1, Predⁿ = 0

2. Cancer patients is detected as Healthy → FN

Risking life

more Harmful →

→ more Harmful →

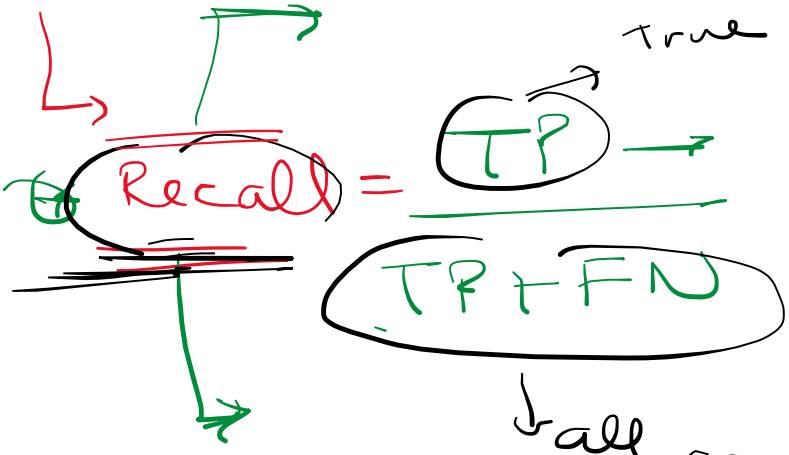


more dangerous

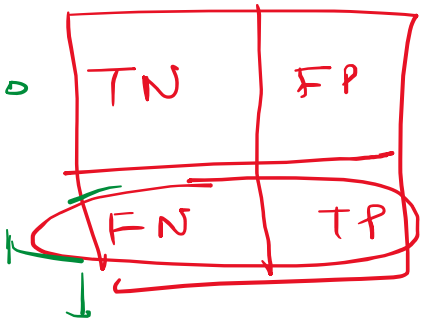
Recall

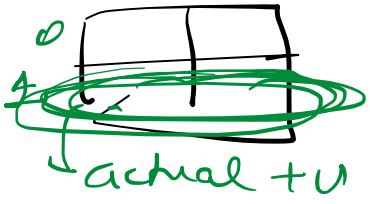
FP

goal ⇒ TP ↑ , min FN ↓



true cancers





Total → 1000 patients

66 → actual cancer patients

Predict → 88 cancer

1 → P
2 → P
3 → 1

$$\text{Recall} = \frac{66}{66} = 1 = 100\%$$

2 → P P
3 → 1 P
× P
1
1000

Recall = 66 / 66 = 1 = 100%

22 → FP

100% Recall

66

CCIN

22 → FP

Conclusion (*)

↓ FN ⇒ (*) Recall

↓ FP → PRECISION

ES

Range = [0, 1]

Dumb model

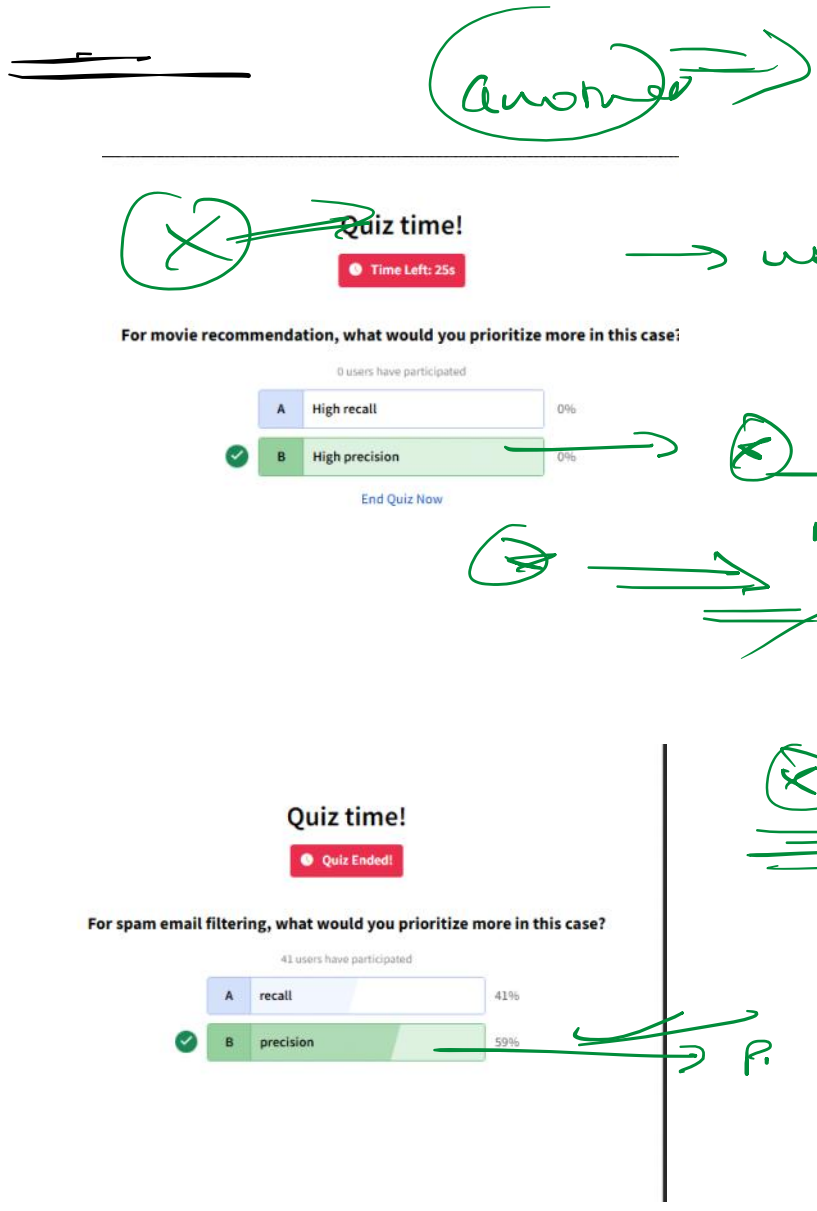
360	0
40	0

Recall = 0 / 40 + 0 = 0

360	0
0	40

Recall = 40 / 40 + 0 = 1

= 100



Fraud Txns

[F] → class 1 → Fraud

[N] → class 0 → not fraud

→ $y=1$, $\hat{y}=0$

→ $y=1$, $y=0$

(1) Predict Txn as legit when it was Fraud. → FN

→ $y=0$, $\hat{y}=1$ ⊗ Financial loss

(2) Predict Txn as Fraud when it was legit → FP

inconvenience ✗ → financial

(3). more dangerous

⇒ Both are imp

FP & FN ↓↓

TP ↑

TP ↑, FP & FN ↓↓

F1 score ⇒

F1 score

(2 × Pre × Re

F1 score

$$\left(\frac{2 \times Pr \times Re\ call}{Pr + Recall}\right)$$

Algos

	Pr	Recall	Avg	F1 score
M_1	0.3	0.8	0.55	0.43
M_2	0.9	0.2	0.55	0.32
M_3	0.7	0.4	0.55	0.5
M_4	0.1	1	0.55	0.18

all same

M_3 ⇒ Best

→ Arithmetic mean ⇒ avg is not right way

AM ⇒ a, b ⇒ $\frac{a+b}{2}$

Harmonic mean ⇒ H.M. = $\frac{2}{\frac{1}{a} + \frac{1}{b}}$

$$HM(P,R) = \frac{2}{\frac{1}{P} + \frac{1}{R}} = \frac{2PR}{P+R}$$

F1 Score

Dumb model → Recall = 0, Prec = 0

$$F1\ Score = \frac{2 \times 0 \times 0}{0+0} = 0$$

ideal model ⇒ F1 Score

$$\frac{2 \times 1 \times 1}{1+1} = 1$$

100%

F1 Score ⇒ [0, 1]

Quiz time!

Time Left: 14s

Why does the F-1 score use Harmonic Mean (HM) instead of Arithmetic Mean (AM) ?

33 users have participated

A	AM penalizes models the most when even Precision and Recall are low.	18%
B	HM penalizes models the most when even Precision and Recall are low.	55%
C	HM penalizes models the most when even Precision and Recall are high.	21%
D	AM penalizes models the most when even Precision and Recall are high.	6%

End Quiz Now

AM

Pr = 1

Recall = 0

AM => $\frac{1+0}{2}$

$= 0.5$

HM = $\frac{2(1)(0)}{1+0} = 0$